

Laboratory 4

TASK: MULTIPROCESSING WITH PIPES AND CONDITIONAL VARIABLE

(use **python** language and **Windows OS**)

1. Create a multiprocessing version of run.py that uses Pipe.
 - Show sending and receiving data via pipe endpoints.
 - Handle bidirectional or unidirectional data flow cleanly(1.5 points)
2. Use condition variable to manage concurrent coordination.
 - Ensure that processes wait for signal/notify events properly.
 - Demonstrate controlled communication.(1 point)
3. Add exception handling and timeout techniques.
 - Wrap send/receive in try/except.
 - Use timeouts where applicable.(0.5 point)
4. Confirm the validity of the result and record the running time performance for comparison with the sequential version (0.5 point)
5. Compare the mean running time and uncertainty with the previous parallel versions (laboratories 2 and 3) and give a recommendation
 - Each parallel should be run 50 times to compute the mean and the uncertainty. Ensure fair comparison, same problem size (20,000,000 or higher), same number of processes (depending on system capacity)
 - Uncertainty is computed as: $\Delta T = \frac{T_{max} - T_{min}}{2}$
 - State which method is better for the problem based on the performance evaluation(1.5 point)

NOTE: Upload the print-screen of final code execution on enauczanie and of course your code in python :)